Problem Statement :**Write a program non-recursive and recursive program to calculate Fibonacci numbers and**
analyze their time and space complexity.

1.Recursive

```
#include<iostream>
using namespace std;
int fibonacci(int n)
{
  if(n<=1)
  return n;
  return fibonacci(n-1)+fibonacci(n-2);
}
int main()
{
  int n;
  cout<<"Enter the number of terms : ";
  cin>>n;
  for(int i=0;i<n;i++)
  {
    cout<<fibonacci(i)<<" ";
  }
  return 0;
}
```

Output:

```
mp-HP-ProDesk-600-G3-SFF:~/Aniket$ ./a.out
Enter the number of terms : 5
0 1 1 2 3 comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ ./a.out
Enter the number of terms : 8
0 1 1 2 3 5 8 13 comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$
```

2.Non Recursive

```
#include<iostream>
using namespace std;

int main()
{
  int n;
  cout<<"Enter the number of terms : ";
  cin>>n;
  int a=0,b=1;
  cout<<"\nFibonacci Series: ";
  for(int i=0;i<n;i++)
  {
    cout<<a<<" ";
    int c=a+b;
    a=b;
    b=c;
  }
  return 0;
}
```

Output:

```
comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ g++ prac11.cpp
comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ ./a.out
Enter the number of terms : 5

Fibonacci Series: 0 1 1 2 3 comp@comp-HP-ProDesk-600-G3-SFF:~/Aniket$ ./a.out
Enter the number of terms : 9

Fibonacci Series: 0 1 1 2 3 5 8 13 21
```